

2022 Ph H2 Q14

Section: Our Dynamic Universe

Topic: Gravitation — Simple Pendulum

A simple pendulum is used to determine g . Period squared T^2 is measured for various lengths L . (a) Draw graph of T^2 vs L . (b) Calculate gradient. (c) Use gradient to find g .

Worked solution

(a) Plot T^2 against L . A straight line through the origin is obtained, showing $T^2 \propto L$.

(b) Gradient = $\Delta T^2 / \Delta L$. From data (best fit):
Gradient $\approx 4.16 \text{ s}^2 \text{ m}^{-1}$

Answer: $4.12 \text{ s}^2 \text{ m}^{-1}$

(c) Equation: $T^2 = (4\pi^2/g)L \Rightarrow \text{gradient} = 4\pi^2/g$.
 $g = 4\pi^2 / \text{gradient} = 39.48 / 4.16 = 9.48 \text{ m s}^{-2}$

Answer: $g \approx 9.6 \text{ m s}^{-2}$

Final answers

(a) Graph: straight line through origin

(b) Gradient $\approx 4.12 \text{ s}^2 \text{ m}^{-1}$

(c) $g \approx 9.6 \text{ m s}^{-2}$

Revision tips

- For a simple pendulum, $T = 2\pi\sqrt{L/g}$.
- Squaring: $T^2 = (4\pi^2/g)L$ gives straight line T^2 vs L .
- Gradient = $4\pi^2/g \Rightarrow g = 4\pi^2/\text{gradient}$.
- Accuracy improved by timing multiple swings and plotting graph.
- Line should pass through origin if no systematic errors.